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A Real-Time Smart Bus Tracking System Using IoT and GPS for Public Transportation Optimization

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Abstract—EvoRoute is an intelligent bus tracking system that enhances public transportation by delivering real-time location updates, precise route details, and a seamless commuter experience. Conventional GPS-based tracking solutions are expensive, prone to inaccuracies, and lack integration with bus schedules. EvoRoute overcomes these limitations with a cost-efficient, reliable, and scalable approach, utilizing real-time GPS tracking and the open-source Leaflet API for mapping. The system provides commuters with live bus location updates, minimizing uncertainty and enhancing travel planning. Its affordability and effectiveness benefit both passengers and transport authorities by improving schedule accuracy and reducing wait times. By offering a user-friendly and accessible solution, EvoRoute redefines public transit with reliable tracking and efficient route management.

Abstract—Keywords—Route Optimization, EvoRouteL, Intelligent Navigation, Pathfinding, Traffic Management.

I. INTRODUCTION

Public transportation systems frequently encounter inefficiencies due to outdated tracking technologies, inconsistent schedules, and a lack of real-time commuter information. EvoRoute is an intelligent bus tracking system designed to improve public transit by delivering real-time location updates, route details, and an enhanced commuting experience. Unlike conventional GPS-based tracking systems that are costly, prone to inaccuracies, and poorly integrated with bus schedules, EvoRoute provides a more affordable and dependable alternative.

Utilizing real-time data, the system accurately tracks bus locations and predicts arrival times, ensuring commuters have reliable information for efficient travel planning. By incorporating Leaflet API for mapping and visualization, EvoRoute

offers an open-source, customizable solution that eliminates the need for expensive alternatives like Google Maps. The system's implementation reduces infrastructure costs while optimizing efficiency, making it a scalable and practical solution for diverse public transportation networks.

With its cost-effectiveness, reliability, and real-time tracking features, EvoRoute revolutionizes the public transit experience. This paper outlines the implementation process, system architecture, and key technological components that enable EvoRoute to function as an efficient and user-friendly tracking system.

II. RELATED WORKS

The advancement of smart transportation systems has led to the development of various solutions aimed at improving bus tracking and optimizing public transit networks. Numerous studies have explored different technologies to enhance the reliability and efficiency of public transportation.

Kamble et al. introduced a system that leverages RFID technology and the ThingSpeak web server to deliver real-time updates on bus locations and seat availability, enhancing commuter convenience in India [1]. Similarly, Singla et al. proposed a GPS-based bus tracking system that integrates historical traffic data to improve the accuracy of bus arrival time predictions. Their system transmits real-time location updates to a central control unit using GPRS, SMS, and RFID technologies [2].

Shree et al. developed a real-time bus tracking system using RFID tags and Arduino controllers, which process RFID data and upload it to the cloud via GSM networks [3]. Kumbhar et al. designed a web-based bus tracking system that employs

Google Maps, GPS, and GPRS to provide real-time location updates every 40 seconds, reducing passenger waiting times [4]. Cedric et al. introduced a Smart School Bus Monitoring System utilizing IoT, RFID, and GPS to ensure student safety during commutes by facilitating seamless communication between parents, schools, and bus operators [5].

Similarly, Sridevi et al. created an IoT-enabled Smart Bus Tracking and Management System that uses Arduino UNO microcontrollers for efficient data storage, allowing commuters to track bus locations in real-time through an Android application [6]. Devarapalli et al. focused on bus arrival time prediction, employing real-time traffic data and queuing theory to enhance accuracy [7]. Rashvand et al. introduced an AI-driven bus arrival prediction system using a Fully Connected Neural Network, which significantly outperformed conventional prediction models [9]. Min et al. advanced real-time traffic forecasting through spatiotemporal correlations, providing predictive insights on traffic patterns up to an hour in advance [10]. Chang et al. developed a dynamic bus trip time prediction model based on Nearest Neighbour Non-Parametric Regression, which was applied to real-world transit data from Seoul. Their approach improved the accuracy of estimated trip durations across different time periods [11]. Garg et al. proposed a GPS-based tracking system that integrates real-time and historical traffic data, enabling commuters to make informed travel decisions [12].

Additional research by Kumbhar et al. introduced a web-based bus tracking system consisting of three key components: a bus module (GPS tracking), a central control unit (data processing), and a user module (interactive web interface) [13]. Jimoh et al. explored a vehicle tracking system based on the Greedy Forwarding Algorithm, improving the efficiency of urban public transportation through wireless technology [14].

Monica et al. implemented a smartphone-integrated GPS-based bus tracking system, delivering real-time updates every 40 seconds to enhance reliability and commuter convenience [15].

These studies highlight various technological approaches in smart transportation, including GPS, RFID, IoT, AI-based prediction models, and real-time data integration. EvoRoute builds upon these advancements by offering a cost-effective, scalable, and real-time bus tracking solution, utilizing Leaflet API for mapping and visualization instead of traditional Google Maps-based approaches.

III. PROPOSED SYSTEM

EvoRoute involves the seamless integration of hardware components, cloud-based data processing, and user-friendly applications to enable real-time bus tracking and predictive analytics. The system is designed to improve public transportation efficiency by providing accurate arrival time predictions and enhancing the overall commuter experience.

A. System Architecture

Fig. 1 is a client-server architecture designed for real-time bus tracking and commuter convenience. It comprises three main components:

- Client Side: A Flutter-based mobile application serves as the user interface, allowing commuters to access live bus location updates and route details. This ensures an intuitive and seamless experience for users.
- Server Side: The backend is built using Flask, which acts as a communication hub between the tracking hardware and the client application. It processes API requests, manages data flow, and ensures smooth interaction between system components.
- Database: MongoDB is used as the primary database to store real-time bus locations, schedules, and user-related information. It enables efficient data management and quick retrieval for enhanced performance.

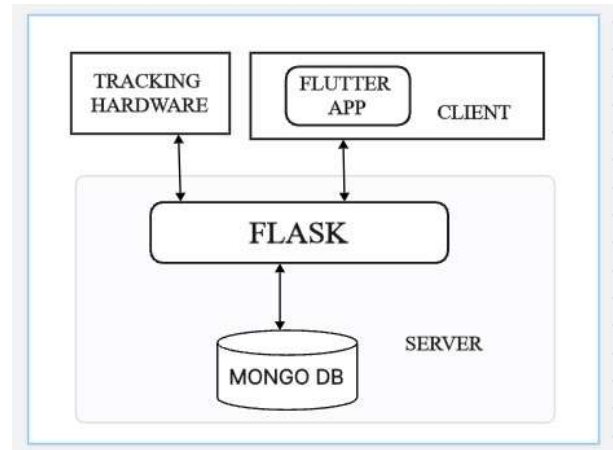


fig. 1 Architecture

B. Hardware Deployment

The hardware configuration of the system includes a NodeMCU (ESP8266/ESP32) microcontroller, a GPS module, and a SIM module, all integrated into KSRTC buses. The GPS module continuously records latitude and longitude coordinates, which are processed by the microcontroller and transmitted to the cloud server using GSM-based communication. This setup enables real-time bus tracking without relying on a Wi-Fi connection.

C. User Interface Development

EvoRoute features a mobile and web-based application for commuters, along with a dedicated admin dashboard for KSRTC officials. The passenger application enables users to monitor bus locations in real time, access estimated arrival times, and receive instant alerts regarding delays, enhancing their travel experience. On the other hand, the admin dashboard equips KSRTC officials with valuable insights into bus operations, occupancy trends, and route efficiency, facilitating informed decision-making. Built using Flutter for mobile platforms, the system ensures a seamless, high-performance, and cross-platform user experience.

D. Real-Time Communication and Updates

EvoRoute leverages Firebase or a cloud-based server to store and synchronize data across multiple devices, ensuring real-time information availability. Bus location updates are

periodically sent to the system's database, allowing commuters to access live tracking details via push notifications. The system utilizes two key data transmission methods: Wi-Fi connectivity and SIM/GPRS-based communication. The Wi-Fi approach enables buses to send location data when connected to available networks, ensuring efficient real-time tracking in areas with stable internet access. On the other hand, the SIM/GPRS method facilitates continuous data transmission via mobile networks, allowing uninterrupted tracking even in regions without Wi-Fi coverage. By integrating these two methods, EvoRoute ensures consistent and reliable bus location updates. This method reduces latency and improves the overall reliability of the system. Predictive analytics is utilized to estimate bus arrival times at designated stops by considering factors such as distance, average speed, and time spent at stops.

Equation to calculate bus fare for all stops:

$$\text{Fare} = \text{minimum fare} + (\text{fare/km} * (\text{total km} - 2.5))$$

IV. HARDWARE

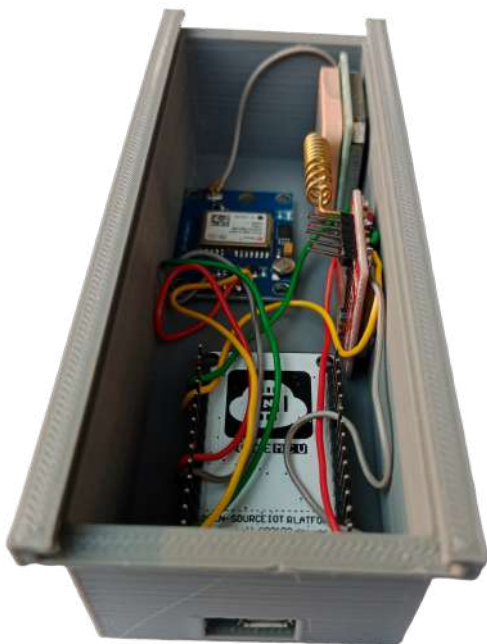


fig.2 Hardware

The EvoRoute system integrates multiple embedded hardware components to facilitate real-time bus tracking and seamless communication. These components function together to collect, process, and transmit data efficiently. Fig. 2 illustrates the hardware setup, showcasing the key elements that enable accurate tracking and reliable data transmission within the system.

A. NodeMCU

A microcontroller equipped with built-in Wi-Fi, serving as the core processing unit for handling data collection and transmission. Manages communication between the GPS module, SIM module, and cloud server.

B. GPS Module

Captures real-time geographical coordinates of KSRTC buses. Continuously sends latitude and longitude data to the microcontroller for processing.

C. SIM Module

Enables GSM-based communication, ensuring seamless data transmission even in areas lacking Wi-Fi connectivity. It facilitates sending bus location data to the cloud and retrieving necessary updates.

D. Wires and Connectors

Comprises essential wiring and connectors for interfacing the hardware components and ensures efficient power distribution and seamless data transmission between modules.

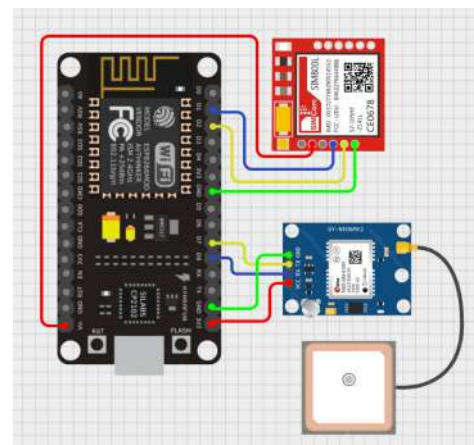


fig.3 Circuit Diagram

Fig. 3 shows the circuit diagram enables real-time bus tracking by leveraging multiple components working in synchronization. The GPS module continuously captures the bus's location data, which is then processed by the NodeMCU microcontroller. This data is transmitted via the SIM module to a cloud-based server, ensuring seamless communication even in areas with limited connectivity. The processed information is then made available through the EvoRoute mobile and web applications, allowing passengers to access real-time tracking updates. This cost-effective and scalable hardware framework ensures reliable performance, low power consumption, and minimal network latency, making EvoRoute an efficient solution for enhancing public transportation systems.

V. SOFTWARE

EvoRoute integrates advanced software technologies to efficiently process, store, and visualize real-time data, creating a seamless experience for users and administrators alike. The system utilizes a combination of cutting-edge tools to ensure accurate bus tracking, optimized scheduling, and a responsive user interface.

A. Frontend (Flutter Framework)

EvoRoute's frontend is developed using the Flutter framework, a cross-platform mobile and web application that ensures users can seamlessly track buses, receive notifications about delays, and plan their journeys.

B. Backend (Flask - Python)

The backend of EvoRoute leverages powerful Python-based frameworks such as Flask or Django to handle server-side logic, process requests from GPS modules, and manage real-time data retrieval.

C. MongoDB

A NoSQL database used to store bus schedules, user preferences, and live tracking data. The flexibility of MongoDB allows the system to handle dynamic data types, making it well-suited for storing real-time location updates and user-specific data.

D. Leaflet API

An open-source mapping solution that displays real-time bus locations and routes, serving as a cost-effective alternative to Google Maps.

VI. RESULTS

The performance of EvoRoute was thoroughly assessed based on key metrics, including real-time tracking accuracy, delay prediction, passenger satisfaction, and the overall impact on KSRTC operations. The system effectively utilized GPS and IoT modules to deliver precise and timely bus location updates with minimal latency, ensuring accurate real-time tracking. Powered by the Leaflet API, the live tracking feature provided users with an intuitive, clear view of bus locations, enhancing the commuter experience. EvoRoute demonstrated significant scalability, with a robust backend infrastructure built using MongoDB and Python frameworks. This infrastructure allowed for efficient data handling and smooth integration of additional buses and routes, facilitating easy expansion as the system grew. The system's GPS-based arrival time predictions significantly outperformed traditional schedule-based estimation methods, leading to more accurate and reliable service.

EvoRoute's administrative dashboard proved to be an invaluable tool in optimizing fleet management. By analyzing real-time data and adjusting bus schedules based on demand, the system improved resource allocation, minimized idle time, and enhanced overall fleet efficiency. This optimization contributed to reduced operational costs and increased service reliability, ultimately improving KSRTC's efficiency in managing public transportation services. The system also highlighted its potential for future scalability by providing flexible solutions for public transport networks of varying sizes, ensuring that both passengers and operators benefit from enhanced service and more efficient operations.

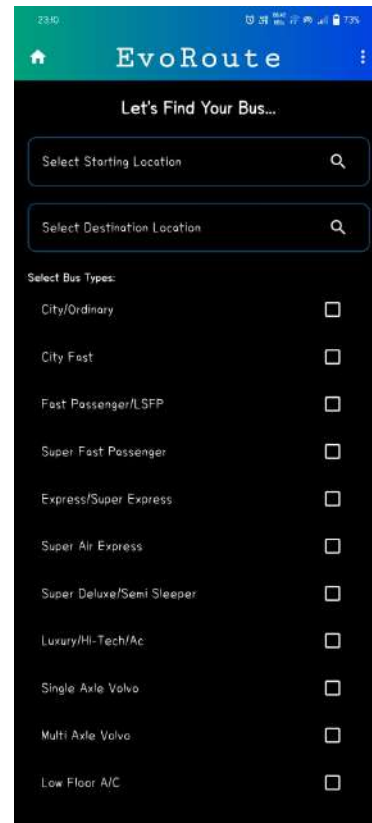


fig.4 Search Bus Page

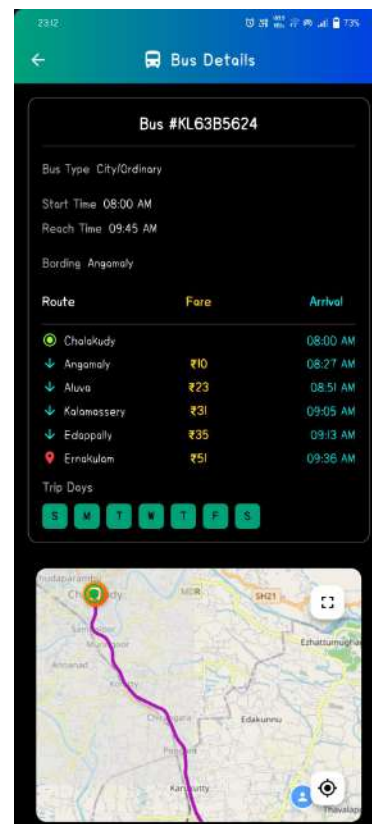


fig.5 Bus Detail page

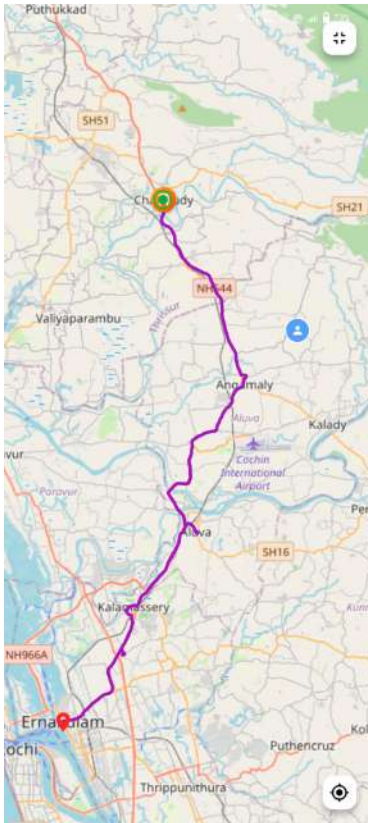


fig.6 Bus Tracking Map

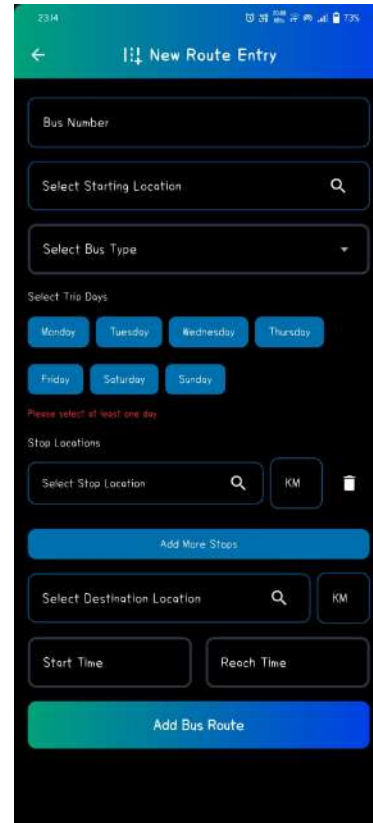


fig.8 Add New Bus Page

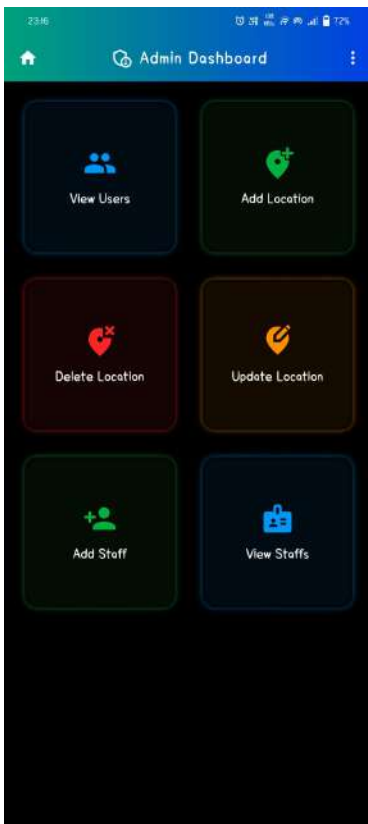


fig.7 Admin page



fig.9 Hardware

Fig. 4 illustrates the bus search interface, allowing users to filter buses based on type and preferred locations. Fig. 5 presents the results page, displaying detailed information about the selected buses according to the user's search criteria. Fig. 6 provides a visual representation of bus locations on an interactive map, enhancing the user experience with a clear geographic overview. Fig. 7 showcases the admin page, enabling administrative control and management of bus-related data. Fig. 8 displays the interface for adding details of new buses, facilitating efficient updates to the system. Lastly, Fig. 9 represents the hardware system responsible for real-time vehicle tracking, ensuring accurate location updates for commuters.

VII. FUTURE ENHANCEMENTS

A. AI-Based Route Optimization

AI-driven Predictive analytics can enhance bus route optimization by evaluating real-time traffic conditions, passenger

demand, and historical travel trends. Utilizing machine learning algorithms, the system can determine the most efficient routes, leading to smoother operations and fewer travel disruptions. Furthermore, dynamic schedule modifications based on real-time data can help minimize delays, improve fleet utilization, and enhance overall service reliability for commuters.

B. Blockchain for Secure Transactions

Integrating blockchain technology can greatly improve the security and transparency of digital transactions by enabling tamper-resistant ticketing and secure payment processing. Utilizing decentralized ledgers helps prevent fraud and unauthorized alterations, ensuring a reliable and verifiable transaction record. Moreover, adopting token-based payments allows for seamless and interoperable fare transactions across various public transportation services, enhancing convenience for commuters.

C. Voice-Assisted and Multi-Language Support

Incorporating voice-assisted technology can improve accessibility for visually impaired passengers by providing bus arrival updates through voice commands, creating a more inclusive commuting experience. Furthermore, expanding language support to include multiple regional languages will accommodate a diverse commuter population, enabling users to interact with the system in their preferred language, thereby enhancing accessibility and user engagement.

D. Predictive Maintenance for Buses

Integrating IoT sensors into buses enables continuous monitoring of vehicle health, allowing the system to detect mechanical issues in real time and schedule predictive maintenance before major failures occur. By leveraging AI-driven diagnostics, the system can analyze sensor data to identify potential malfunctions and provide early alerts to KSRTC, helping prevent unexpected breakdowns, minimize service disruptions, and enhance overall fleet reliability.

VIII. CONCLUSION

EvoRoute is an innovative and integrated solution aimed at enhancing the efficiency, reliability, and accessibility of public transportation. By incorporating real-time GPS tracking and delay predictions, EvoRoute effectively addresses critical challenges such as inconsistent bus schedules, extended wait times, and inefficient fleet management within KSRTC. This system significantly improves the overall commuting experience while optimizing transit operations for transport authorities.

EvoRoute utilizes advanced data processing techniques to deliver accurate, real-time updates on bus locations, estimated arrival times, and route changes. By leveraging predictive analytics, the system enhances journey planning by helping passengers make informed decisions based on live data. The mobile application, designed with a user-friendly interface, ensures seamless access to essential travel information, allowing commuters to track buses, receive notifications, and plan their trips with ease. For transport authorities, EvoRoute provides

critical operational insights, enabling better route planning, scheduling adjustments, and resource allocation to improve service efficiency.

A key strength of EvoRoute lies in its scalability and adaptability. The system efficiently handles vast amounts of real-time data through its robust backend infrastructure, built using Python framework Flask, and a NoSQL MongoDB database for storing and managing bus schedules, user preferences, and live tracking data. The integration of Leaflet API, an open-source mapping solution, allows EvoRoute to provide a cost-effective alternative to proprietary mapping platforms while ensuring accurate route visualization and vehicle tracking.

REFERENCES

- [1] Kamble, P. A., Vatti, R. A. (2017, December). Bus tracking and monitoring using RFID. In 2017 Fourth International Conference on Image Information Processing (ICIIP) (pp. 1-6). IEEE.
- [2] Singla, L., Bhatia, P. (2015, September). GPS based bus tracking system. In 2015 International Conference on Computer, Communication and Control (IC4) (pp. 1-6). IEEE.
- [3] Shree, A. D., Anusuya, J., Malathy, S. (2019, March). Real time bus tracking and location updation system. In 2019 5th International Conference on Advanced Computing Communication Systems(ICACCS) (pp. 242-245). IEEE.
- [4] Kumbhar, M., Survase, M., Mastud, P., Salunke, A., Sirdeshpande, S. (2016). Real time web based bus tracking system. International Research Journal of Engineering and Technology, 3(2), 632-635.
- [5] Cedric, U. M., U. H. I. R. E., Mbanzabugabo, J. B. (2022). Smart School Bus Monitoring and Notification System Using Rfid and Gps.
- [6] Sridevi, K., Jeevitha, A., Kavitha, K., Sathya, K., Narmadha, K. (2017). Smart bus tracking and management system using IoT. Asian Journal of Applied Science and Technology (AJAST) Volume, 1.
- [7] Devarapalli, H. N. R. K. (2015). Predicting Bus Arrival Time based on Traffic Modelling and Real-time Delay. Int.J. of Eng. Research and Technol.(IJERT), 4(6), 421-426.
- [8] Sharif, S. A., Suhaimi, M. S., Jamal, N. N., Riadz, I. K., Amran, I. F., Jawawi, D. N. (2018, July). Real-Time campus university bus tracking mobile application. In 2018 Seventh ICT International Student Project Conference (ICT-ISPC) (pp. 1-6). IEEE.
- [9] Rashvand, N., Hosseini, S. S., Azarbayjani, M., Tabkhi, H. (2023). Real-Time Bus Arrival Prediction: A Deep Learning Approach for Enhanced Urban Mobility. arXiv preprint arXiv:2303.15495.
- [10] Min, W., Wynter, L. (2011). Real-time road traffic prediction with spatio-temporal correlations. Transportation Research Part C: Emerging Technologies, 19(4), 606-616.
- [11] Chang, H., Park, D., Lee, S., Lee, H., Baek, S. (2010). Dynamic multi-interval bus travel time prediction using bus transit data. Transportmetrica, 6(1), 19-38.
- [12] Garg, N., Gawande, P. V., Kurekar, P. P., Kharat, D. B. (2017). Bus tracking using gps and real time prediction. International Research Journal of Engineering and Technology, 4(4), 3477-347.
- [13] Kumbhar, M., Survase, M., Mastud, P., Salunke, A., Sirdeshpande, S. (2016). Real time web based bus tracking system. International Research Journal of Engineering and Technology, 3(2), 632-635.
- [14] Jimoh, O. D., Ajao, L. A., Adeleke, O. O., Kolo, S. S. (2020). A vehicle tracking system using greedy forwarding algorithms for public transportation in urban arterial. IEEE Access, 8, 191706-191725.
- [15] Monica, K. I., Gurupriya, S., Magdaline, S. A. (2019). Bus Tracking System using GPS on Smartphones. International Journal of Engineering Research Technology (Ahmedabad), 7(11), 1-4.